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Analysis of climbing postures and movements in sport climbing for realistic 3D climbing animations

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Abstract

Recently, sport climbing has become popular throughout the world. Sport climbing is an extreme sport where people climb an artificial wall with various artificial holds using their hands and feet. Technique training related to using hands and feet is necessary for novice climbers, hence we need effective training contents. The purpose of this research is to analyze postures and movements in sport climbing using Microsoft Kinect for realistic 3D climbing animations. A professional male climber participated in this research in order to obtain good movements for the animations. We divided a climbing cycle into four phase and analyzed each phase. The posture and movement data in sport climbing which are analyzed through this research will be used to make realistic 3D character animation.

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1. Introduction

Sport climbing is an extreme sport where people climb an artificial wall with various artificial holds using their hands and feet. Recently, sport climbing has become popular throughout the world. According to the International Federation of Sport Climbing, 25 million people (over 2 million Europeans) enjoy sport climbing regularly. There

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are more than 300 indoor climbing gyms in London and Paris. The number of indoor climbing gyms increased more than 50% from 2007 to 2012.

There are innumerable climbing methods and techniques. But in most of climbing movements, people climb an artificial wall using the three-holds-rule and the Left-Right rule [1]. Three-holds-rule is that a climber climbs a wall using three holds. It means that if a climber uses three holds the climber can maintain balance on a wall. The Left-Right rule is that a climber climbs a wall using the contralateral hand-foot in pairs. If the climber climbs using the ipsilateral hand-foot in pairs, the climber will swing uncontrollably out from the wall, which causes an action known as the barn-door. Thus, techniques related to hands and feet are the most important techniques for effective climbing. Novice climbers need to learn these techniques through the lessons from climbing instructors. Climbing instructors feel inconvenient when they have to demonstrate the same climbing techniques in many lessons. For this reason, it is necessary to provide climbing animations with which novice climbers can train fundamental climbing techniques easily. But when we make character animations without the analysis of sport climbing postures and movements, it is hard to generate sport climbing postures and movements including realistic velocity and angles. In order to make effective climbing animations, climbing postures and movements must be studied in advance. They have been studied in terms of kinematics, kinetics, muscle operating ranges, and energetics [2], especially common patterns and differences between various climbers [3].

There are marker-based systems and marker-less systems in motion capture methods for motion analysis. Marker-based motion capture systems make it possible to measure accurately, but there are limitations due to the high cost as well as using many cameras and attaching markers to the body. For these limitations, marker-less motion capture systems have been developed recently. Microsoft Kinect is a marker-less motion capture devices. Kinect is at very low cost compared with marker-based motion capture systems. Although the accuracy of Kinect is inferior compared to marker-based devices, Kinect has been proven to be useful in the measurement of knee and hip angles [4], or elbow and shoulder angles [5].

The purpose of this research is to analyze fundamental climbing postures and movements in sport climbing using Microsoft Kinect for realistic 3D climbing animation.

2. Methods

2.1. Data Collection

A professional male climber participated in this research (height: 166 cm, weight: 54kg, age: 33, climbing career: 20 years, former Korean national athlete). The climber wear slim clothes and avoid shiny fabrics to reduce noise in the resulting data. For the analysis of climbing postures and movements in sport climbing, RGB videos and depth data are recorded from two Kinects (Kinect for Windows v2, Microsoft) and iPi Recorder (iPi Recorder 3, iPi Soft LLC). iPi Recorder is a software for capturing, playing back and processing videos obtained from multiple cameras and depth sensors. Captured records can be used for motion tracking in iPi Mocap Studio. First, the two Kinects are installed in front of a clear acrylic climbing wall, in accordance with recommendations from the iPi Soft LLC. By recording with a clear acrylic climbing wall, it is possible to obtain fine depth data. When using a regular wall, hands or feet close to the wall can't be recognized. Next, a background is recorded. Background evaluation is needed before starting to record. The background is a picture of only static objects in the recording area. Evaluated background data is used later to easily detect moving objects. After, a calibration video is recorded to compute accurate Kinect positions and orientations. Calibration is required for multi-Kinect system setup. For calibration, we used a white foam board (horizontal size: 61 cm, vertical size: 91 cm) and moved the board back and forth, with slight side-to-side rotation.

The recording procedure was fully explained to the climber. And the climber entered the recording area and took a preparation pose before the climbing. Then, the climber climbed the clear acrylic wall so that fundamental postures and horizontal movements of sport climbing could be measured. The distance between the holds was about one meter, so that the climber could reach the holds easily. The climber climbed the wall using his climbing style and speed. The climber repeated horizontal movements 30 times.

2.2. Data analysis

To make the 3D skeleton data of the climber from the recorded depth video, we use iPi Mocap Studio (iPi Mocap Studio 3, iPi Soft LLC). iPi Mocap Studio is a software for tracking an actor's motion by analyzing multi-camera(or depth sensor) video recordings. iPi Mocap Studio automatically track the body of the climber in the depth video. However, a 'clean-up' process is necessary because of tracking errors. Tracking errors in a few depth video frames may spread to multiple subsequent frames, resulting in tracking gaps. Tracking errors are caused by occlusion, indistinctive pose, or very fast motion with motion blur. To clean-up tracking gaps, the tracking results are edited manually using editing tools (Rotate, Move and Inverse Kinematics) of the software. Next, post-processing filters are used for the tracked skeleton; a jitter removal filter and a trajectory filter. The jitter removal filter can remove unexpected noises, and can preserved sharp, dynamic movement. The trajectory filter's purpose is to filter out minor noises that remain after applying the jitter removal filter. After post-processing, the 3D skeleton data of the climber is exported into EXCEL and BVH file formats.

To analyze the exported 3D skeleton data, we use a public domain software (BVH Viewer, Sheffield Hallam University). We use BVH Viewer since it is made in accordance with International Society of Biomechanics guidelines. For example, elbow flexion-extension is calculated by first Euler angle (ZXY convention) and shoulder angle is calculated by second Euler angle (YXY convention). [5, 6, 7, 8]

3. Result

The limb velocity and joint angles during climbing were calculated using Microsoft Kinect, iPi Recorder & Mocap Studio, and BVH Viewer. A climbing cycle moving from a hold to another one was divided into the initial posture and Phase 1~4, based on the movement velocity change. Figure 1 shows the velocity change of hip, forearms, shoulders, thighs, and shins.

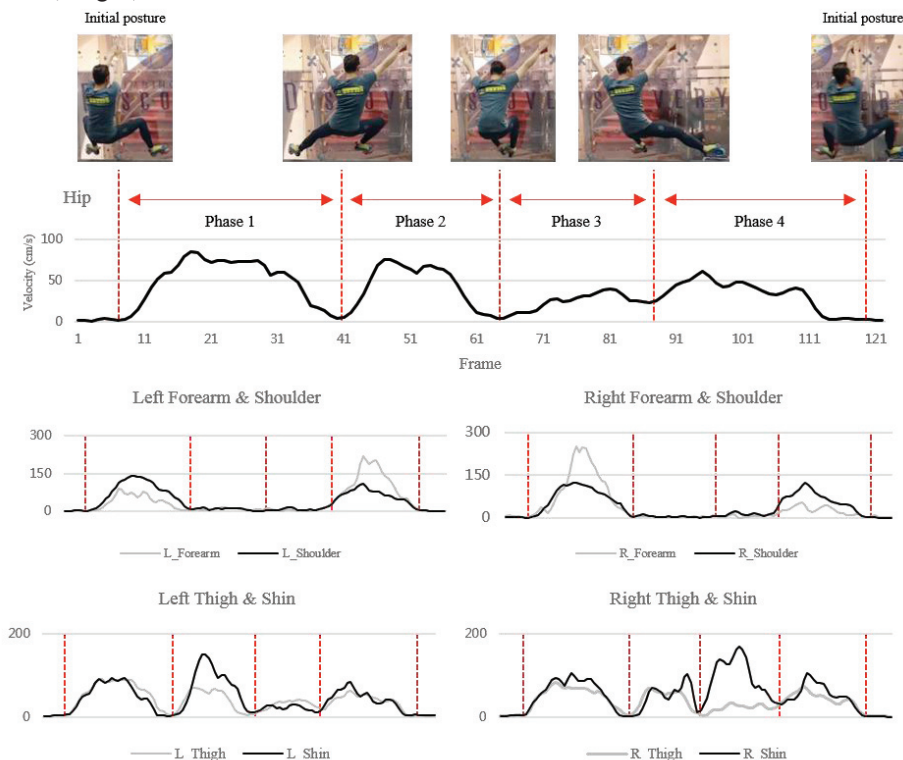


Fig. 1. The movement velocity variation of forearms, shoulders, thighs and shins in trial #7 (the movement of the right direction).

Each phase was defined from the time the segmented movement begins to the time the movement finishes, based on velocity change. In Fig. 1, Phase 1 starts when the velocity of the right forearm and shoulder (the moving direction hand) begins to increase and ends when the velocity decreases to its minimum. Phase 2 starts when the velocity of the left thigh and shin (the contralateral foot) begins to increase and ends when this velocity decreases to its minimum. Phase 3 starts when the velocity of the right thigh and shin (the moving direction foot) begins to increase and ends when this velocity decreases to its minimum. Phase 4 starts when the velocity of the left forearm and shoulder (the contralateral hand) begins to increase and ends when this velocity decreases to its minimum.

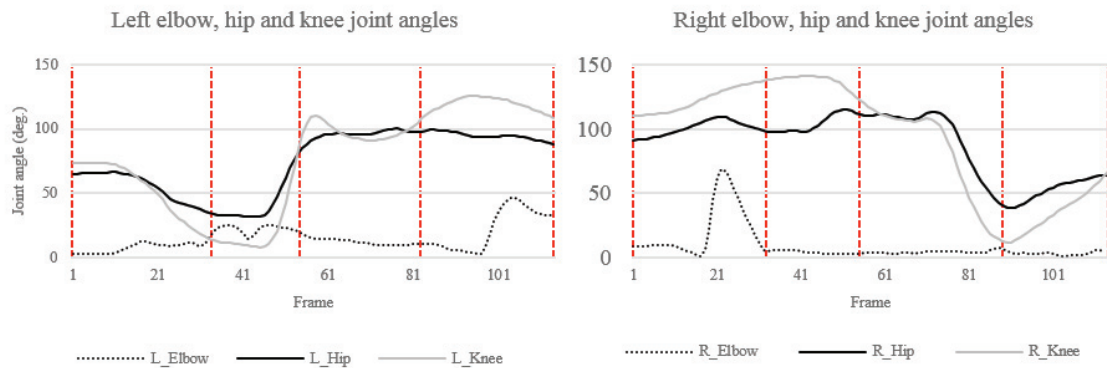


Fig. 2. The joint angles variation in trial #21 (the movement of the right direction).

Table 1. The joint flexion angles of the finished posture of each phase (the movement of the right direction, n=30).

Phases and Joints		Mean ($\pm$ SD) (deg)	Min (deg)	Max (deg)	Phases and Joints		Mean ($\pm$ SD) (deg)	Min (deg)	Max (deg)
Initial posture	Left elbow	3.88 ($\pm$ 2.85)	0.18	10.27	Finished posture of Phase 1	Left elbow	11.95 ($\pm$ 7.63)	2.23	25.34
	Right elbow	9.255 ($\pm$ 4.13)	2.01	17.06		Right elbow	3.84 ($\pm$ 2.32)	0.12	10.27
	Left hip	69.51 ($\pm$ 5.51)	58.14	77.45		Left hip	26.37 ($\pm$ 7.35)	0.90	39.07
	Right hip	79.85 ($\pm$ 16.61)	54.20	107.62		Right hip	99.86 ($\pm$ 8.11)	82.56	121.04
	Left knee	79.49 ($\pm$ 8.02)	63.10	96.28		Left knee	8.84 ($\pm$ 2.73)	4.51	14.49
	Right knee	97.32 ($\pm$ 10.60)	75.22	113.70		Right knee	142.62 ($\pm$ 5.38)	133.14	158.78
Finished posture of Phase 2	Left elbow	8.91 ($\pm$ 4.11)	0.77	17.05	Finished posture of Phase 3	Left elbow	2.95 ($\pm$ 2.44)	0.03	10.10
	Right elbow	2.10 ($\pm$ 1.04)	0.16	4.15		Right elbow	3.43 ($\pm$ 2.28)	0.20	8.11
	Left hip	90.95 ($\pm$ 4.04)	83.72	102.44		Left hip	94.08 ($\pm$ 4.24)	86.29	103.77
	Right hip	111.69 ($\pm$ 7.12)	98.13	122.83		Right hip	39.33 ($\pm$ 4.47)	28.40	50.83
	Left knee	105.89 ($\pm$ 5.72)	90.79	117.72		Left knee	127.14 ($\pm$ 3.01)	120.63	134.31
	Right knee	177.71 ($\pm$ 7.53)	100.07	128.18		Right knee	15.29 ($\pm$ 3.96)	4.73	22.30

Figure 2 shows the joint angles change of hip, elbows, and knee. Table 1 shows mean ($\pm$ SD), minimum, and maximum values of the joint flexion angles for the finished posture of each phase (n=30).

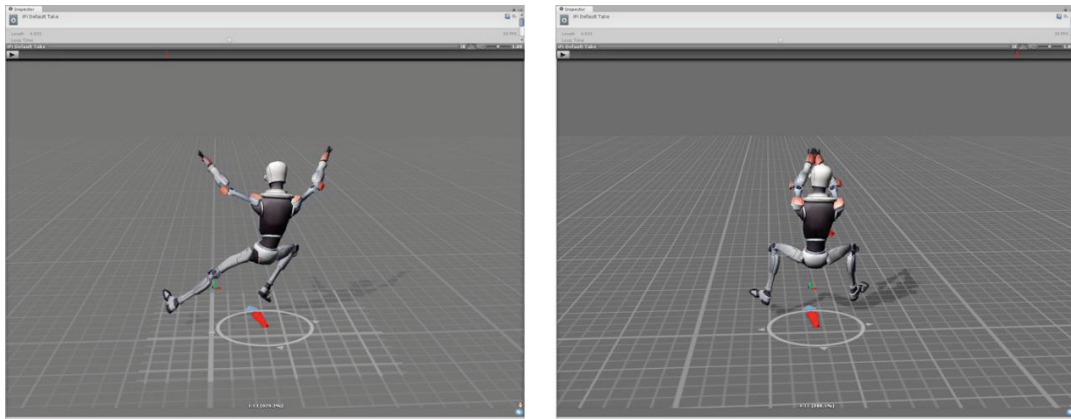


Fig. 3. Generated 3D character animation.

Figure 3 shows the 3D character animation generated through the motion capture.

4. Discussion

We analyzed a climbing cycle using obtained skeleton data. There are differences between each phase in the velocity of each limb and the range of each joint. These differences must be considered to make realistic 3D character animations. The velocity variation of each limb in each phase must be reflected in 3D character movements. The generation of unnatural climbing postures and movements can be minimized by restricting the range of 3D character motions based on the analyzed data.

According to previous research [9], the skeleton tracking of Kinect is not so accurate for recognizing hands and feet when close to the wall. To enhance the Kinect tracking accuracy, we used a clear acrylic wall. Since the infrared light of the depth camera penetrate the clear acrylic wall, we can obtain precise motion capture data.

There are some limitations in the research. First, even though tracking accuracy was enhanced by using a clear acrylic wall, tracking errors still occurred. Because of that, we cleaned up tracking errors using iPi software editing tools and post-processing filters. There might be slight errors (e.g. angle, rotation) in the clean-up tracking process. Future research is necessary to enhance the tracking accuracy of a climber using Kinect or other methods. Second, a professional climber participated in this research. To understanding of sport climbing postures and movements, it is necessary to analyze various climber's postures and movements (e.g. athlete, novice) in future research.

5. Conclusion

In this research, we obtained motion capture data and analyzed sport climbing postures and movements using Kincet, iPi Recorder & Mocap Studio, and BVH Viewer.

3D character animations are generated by processing motion capture data (Fig .3). We can observe climbing postures and movements from various viewpoints through these animations. These animations will become useful training contents for novice climbers.

As future works, we will generate basic 3D character animations using inverse kinematics (IK) method. IK method can move the whole 3D human model to desired position and orientation, while generating proper angles for the joints from the given hands and feet positions. The reality of the animation can be improved by applying the analyzed data in this research. Through the process mentioned above, we will be able to implement real time animations showing postures and movements optimized for virtual route set by the users.

Acknowledgements

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